EU INNOVATION HUB

ANNUAL REPORT 2024

EU INNOVATION HUB FOR INTERNAL SECURITY ANNUAL REPORT

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The EU Innovation Hub for Internal Security Annual Report

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EU INNOVATION HUB FOR INTERNAL SECURITY

EU Innovation Hub for Internal Security
Annual report 2024

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1. Introduction

This report provides a review of the progress achieved by the EU Innovation Hub for Internal Security (hereinafter, the Innovation Hub or the Hub) in 2024. The report was prepared by the Innovation Hub Team and is addressed to the Council's Standing Committee on Operational Cooperation on Internal Security (COSI) following endorsement by the Innovation Hub's Steering Group. This document is a retrospective report on the achievements of 2024; a description of the activities planned for the timeframe 2023-2026 can be found in a separate document entitled *EU Innovation Hub for Internal Security – Multi-annual planning of activities 2023-2026*, including a detailed *Draft Work Plan 2025 – Addendum to the Multi-annual planning of activities 2023-2026*.

Following the outcome of the Hub Steering Group meeting of 4 October 2023, the Hub Team carried out its activities in four thematic clusters:

- Cluster 1: Foresight and key enabling technologies
- Cluster 2: Biometric recognition systems: data quality, evaluation and standardisation
- Cluster 3: Artificial intelligence
- Cluster 4: Encryption & access to data

A thematic cluster is defined as a thematic priority for the Hub in which individual Hub Team members agree to work collaboratively, with one of them volunteering as a leader. Cluster leaders steer the work in that area while respecting the autonomy of the Hub members carrying out activities within the cluster. They report on the progress and results of the activities to the entire Hub Team on a regular basis. The cluster approach is also reflected in the structure of this report. In addition to the work in the thematic clusters, the Hub Team continued to work jointly on other topics of interest related to innovation in the field of EU internal security, as well as on events.

2. The Innovation Hub's mandate

In line with the JHA Council outcome of October 2019, the EU Innovation Hub for Internal Security was established by COSI on 18 February 2020. The Innovation Hub is a collaborative network of innovation labs. It is a cross-sectorial EU platform aimed at ensuring coordination and collaboration between all stakeholders in the broader field of internal security. It was set up to support the delivery of innovative cutting-edge products for citizens' security in the area of freedom, security and justice. The objective of the Innovation Hub is to contribute to establishing a common innovation landscape for internal security and to promote the alignment of innovation and security research efforts across Europe.

Under the guidance of the Hub's Steering Group, the Innovation Hub's work is coordinated by the Hub Team, which serves as the operational arm of the Hub. The Hub Team is composed of representatives from the

innovation labs or contact points of the JHA Agencies – European Union Agency for Law Enforcement Training (CEPOL), European Union Agency for Asylum (EUAA), European Union Drugs Agency (EUDA), European Union Agency for the Operational Management of Large-Scale IT Systems (eu-LISA), European Union Agency for Criminal Justice Cooperation (Eurojust), European Union Agency for Law Enforcement Cooperation (Europol), European Union Agency for Fundamental Rights (FRA) and European Border and Coast Guard Agency (Frontex), the European Public Prosecutor's Office (EPPO), the Commission (DG HOME and DG Joint Research Centre (JRC)), and the General Secretariat of the Council (DG JAI and the Office of the EU Counter Terrorism Coordinator (EU CTC)).

The decision by the European Public Prosecutor's Office to join the Hub, approved by the Steering Group in November 2024, serves as clear evidence of the Hub's effectiveness in delivering on its mandate. The growing interest from new members such as the EPPO demonstrates not only the Hub's success in achieving its objectives but also its strong appeal as a key platform for exchanging knowledge and fostering innovation.

3. The Innovation Hub's resources

Throughout 2024, the resources available for the work on the Innovation Hub remained at similar levels as in 2023. The Innovation Hub Secretariat at Europol in The Hague was composed of three Europol staff members who were partly assigned to the Hub and two representatives seconded to the Hub Secretariat from eu-LISA (50%) and the French Police (50%). In 2024, eu-LISA remained the only agency that seconded a staff member to strengthen the Hub Secretariat, as requested by COSI in 2020. Notable contributions were also made by the leaders of the four thematic clusters (DG HOME, eu-LISA, Europol and JRC) who organised meetings and propelled the activities of the clusters as outlined in the Work Plan 2024.

The activities and work described in this Annual report could not have been accomplished without the contributions of the Hub Team members, who dedicated their time to participate in the regular Hub Team and Cluster meetings, contributed to particular projects, and reviewed and provided comments on documents and reports produced by the Hub Secretariat.

4. Summary of 2024 work – Basic tasks

Despite the limited resources available to the Innovation Hub throughout 2024, it carried out a number of activities in response to the priorities identified by COSI, and initiated new activities in the thematic clusters, which are described in this report.

4.1 Annual event

One of the tasks indicated by COSI was the organisation of an annual Hub event. The Innovation Hub Annual Event, funded by the Commission and organised in collaboration

with the CERIS community, took place in Brussels on 28 November 2024. Attendance was by invitation only, with a focus on representatives of national authorities (including COSI representatives), academic experts and practitioners operating in the area of internal security. In total, more than 180 participants attended the in-person event.

The event featured keynote speeches delivered by Mr Bartjan Wegter, EU Counter-terrorism Coordinator, and Olivier Onidi, Deputy Director General of DG Migration and Home Affairs of the European Commission. There was also a series of panel discussions coordinated by the Hub's thematic clusters, focusing on topics pertinent to each cluster, namely:

- *Foresight as a means to better prepare for emerging security challenges*
- *Advancing biometric technology and identity management in the EU: exploring the need for testbeds and sandboxes*
- *The benefits of explainable AI in internal security*
- *Encryption and access to data*

4.2 Hub collaboration platform

Among the other tasks assigned by COSI, the Hub was tasked with setting up a common platform for exchanging information and disseminating activities among participating innovation labs and other relevant stakeholders.

The Hub Team initiated a discussion with CEPOL on the potential use of its Law Enforcement Education (LEEd) platform; this was based on the mapping of existing platforms carried out by the Hub Team in 2023, as a first step, and in order to address the need for a platform to support cooperation and information exchange within the EU Innovation Hub and with other relevant stakeholders. The results of this discussion were presented at the Hub Steering Group meeting on 21 December 2023. The proposal to use LEEd as a solution to meet the needs of the Hub Team in the short to medium term was positively received by the Steering Group.

However, since CEPOL IT infrastructure was affected by a cybersecurity incident in early 2024, the implementation of the knowledge and collaboration platform for the Hub, using CEPOL's LEEd, has been postponed until the necessary actions are taken to allow the work to continue in early 2025.

4.3 Steering Group and Hub Team meetings

As already mentioned, in 2024 the Hub introduced a new approach to work based on thematic clusters. Therefore, most of the activities and projects that took place in 2024 were coordinated by the individual clusters. As a result of this change, the scope of work of the Hub Team has changed: it now focuses primarily on the overall coordination of the

Hub activities and information sharing, and on preparation for the meetings and documentation for the Hub's Steering Group. Throughout 2024, the Hub Team met on average every two to three weeks, whereas the Steering Group met three times during the year. Two of the Steering Group meetings were formal (online meetings on 14 March and 11 October) and one was informal, with the participation of the Hub Team (in-person meeting on 29 November following the Innovation Hub Annual Event in Brussels).

During the regular Hub Team meetings, members of the team provided updates on their ongoing innovation activities, including: the results of a project focusing on key enabling technologies and privacy enhancing technologies for the JHA domain, coordinated by Frontex; the CEPOL Research Week; eu-LISA Industry Roundtable events; Europol Industry and Research Days; Frontex Conference on Future Training Technologies; and the work of the High-Level Group on Access to Data for Effective Law Enforcement. On two occasions, the Hub Team welcomed representatives of the Belgian and Hungarian Presidencies of the Council of the EU, who presented their priorities in the area of internal security, specifically focusing on innovation. The Hub Team also received presentations delivered by research and innovation projects funded by the EU, including AHEAD and ENACT on foresight, and project demos by VIGILANT and FERMI on countering fake news and disinformation.

4.4 Activities in cooperation with the Presidency of the Council

During the Belgian Presidency of the Council, COSI took note of the Innovation Hub's reports (annual report 2023, report on the 2023 annual event, and work plan for 2024), welcomed the Innovation Hub's regular reporting to COSI, encouraged the inclusion of Innovation-Hub-related activities in the JHA agencies' financial programmes and endorsed the possible organisation of a high-level strategic reflection on the future of EU security research and innovation. In April 2024, COSI discussed the impact of quantum technologies on internal security. Delegates considered this discussion to be timely and invited the Innovation Hub to raise awareness about quantum technologies with the support of the respective agencies, and to establish cooperation with industry and academia. Europol was called upon to continue its activities in quantum technologies with the European Clearing Board.

The Hungarian Presidency of the Council promoted innovation in internal security during its term, making best use of the reinforced thematic work streams of the Innovation Hub. With the objective of raising awareness about major aspects of innovation in internal security, the Presidency presented the deliverables and ongoing activities of the Hub at Council working party meetings. Following an exchange of views on encryption at the COSI Support Group meeting on 9 September 2024, the Hub's work on artificial

intelligence was considered at the Working Party on JHA Information Exchange (IXIM) meeting on 11 September 2024, its work on biometrics at the IXIM meeting on 18 October 2024, and its work on foresight at the IXIM meeting on 5 December 2024. Throughout all these discussions, Member States appreciated the work done by the thematic clusters of the Hub, able to be delivered thanks to the commitment of its members. The Presidency informed COSI about the outcome of these thematic discussions on 27 November 2024 and invited the Hub to take it into account in its further planning. The Presidency, the Hub and the European Clearing Board (EuCB) organised a workshop 'Impact of Quantum Technology on EU Law Enforcement Authorities' in Brussels on 27 November 2024.

5. Activities of the thematic clusters

5.1 Cluster 1: Foresight & key enabling technologies

Cluster leader: DG HOME, European Commission

Cluster members : CEPOL, DG JRC, EUDA, eu-LISA, Europol, Frontex, GSC

This cluster was formally launched with a kick-off meeting on 15 February 2024. In addition to the kick-off meeting, the Foresight Cluster organised a joint meeting with the EU-ANSA[1] Futures Cluster during the CEPOL Research Week in June in Budapest, and an online cluster meeting in October 2024. The cluster also contributed to the organisation of a CERIS[2] workshop on foresight and key enabling technologies, which took place in Brussels on 5 March 2024. The workshop brought together participants from different security sectors, including border management, customs and law enforcement (among which security practitioners, researchers, industry and civil society organisations), and representatives from the European Commission and NATO.

Throughout 2024, the Foresight Cluster was involved in a number of projects. First, the cluster closely collaborated with the AHEAD project, one of the key outcomes of which was the assessment of the foresight roadmaps developed by the project in the areas of

[1] EU Agencies Network on Scientific Advice

[2] Community for European Research and Innovation in Security

fighting crime and terrorism and border management. Second, it contributed to the mapping of key enabling technologies carried out by Frontex. Finally, the cluster contributed to technology scanning and analysis of key enabling technologies, which will result in two case studies:

- Case study 1: *EUDA study on technologies related to drug markets*
- Case Study 2: *JRC study on emerging security challenges in the EU stemming from key enabling technologies*

Finally, the Foresight Cluster also contributed to the organisation of the Hub Annual Event by organising a panel discussion on foresight as a means to better prepare for emerging security challenges.

5.2 Cluster 2: Biometric recognition systems: data quality, evaluation & standardisation

Cluster leader: eu-LISA

Cluster members: DG HOME, DG JRC, Europol, FRA, Frontex

This cluster was formally launched with a kick-off meeting on 26 February 2024. In addition to the kick-off meeting, the Biometrics Cluster organised an online cluster meeting in June and an in-person meeting in September 2024, back-to-back with the Research Projects Conference organised by the European Association for Biometrics in Darmstadt, Germany.

Participants of the cluster shared information about relevant projects carried out by the members, including:

- The evaluation of the shared biometric matching services carried out in collaboration between eu-LISA and DG JRC.
- The App4EES project, in which Frontex piloted an application for pre-enrolment of biographical and biometric data in the Entry/Exit System with the help of a mobile application. The open-source face image quality tool, developed by the German Federal Office for Information Security and hosted by eu-LISA since mid-2024, was used in this project to assess the quality of facial images acquired using the app.
- The project focusing on the fundamental rights implications of the use of remote biometric identification for law enforcement purposes initiated in 2024 by the Fundamental Rights Agency.

The Biometrics Cluster also contributed to the organisation of the Hub Annual Event by organising a panel discussion focusing on the evaluation of biometric recognition systems and the related need for testbeds and sandboxes in the EU.

5.3 Cluster 3: Artificial Intelligence

Cluster leader: Europol

Cluster members: CEPOL, CTC, DG HOME, DG JRC, EPPO, EUAA, EUDA, eu-LISA, Eurojust, FRA, Frontex, GSC

The AI Cluster launched its activities with an online meeting on 1 March 2024 outlining its planned activities, in particular those supporting the implementation of the EU AI Act, including through CC4AI, and the Commission's EU Security Data Space for Innovation (EU SDSI).

The kick-off meeting was followed up by an in-person workshop organised by Europol on 8-9 April in The Hague, where participants discussed AI governance and the implementation of the AI Act, and bias and explainability in artificial intelligence systems. The meeting also included demonstrations of a number of projects and AI tools and solutions presented both by members of the cluster and external speakers, and an update on the development of Europol's research and innovation sandbox[3], as a central element of the EU SDSI.

In addition to the above, the AI Cluster organised an in-person meeting in Budapest during the CEPOL Research Week in June. This meeting focused on the collection of feedback from Hub members on challenges related to the implementation of the AI Act, which was provided to DG HOME for onward transmission to the Commission's new AI Office. The meeting furthermore saw AI tool demos, including on AI use in research and publication, such as RAGs (Retrieval-Augmented Generation techniques).

The cluster members met in hybrid format in The Hague on 4 September, where participants continued to discuss challenges associated with the implementation of the EU AI Act and saw a presentation of Europol's 'AI and policing' report.

The last meeting of the AI Cluster took place on 17 December 2024 online. Among the topics discussed were the progress of the TESSERA project supporting the EU SDSI, and the delivery of the Europol research and innovation sandbox renamed ODIN (Operational Data for Innovation) in December 2024.

Throughout the year, cluster participants focused primarily on the topic of AI governance and the implementation of the EU AI Act. In particular, they provided feedback to the Commission on the interpretation and implications of the AI Act for the internal security community, shared best practices on ensuring regulatory compliance and the rollout of AI-based technologies, and exchanged information on the outcomes of one of the Hub's flagship projects – Accountability Principles for AI (AP4AI) – in particular focusing on the compliance checker for AI (CC4AI) tool. Other areas of focus of the

[3] Implementing art. 33a Europol Regulation

cluster's work were tackling bias in AI systems and explainability in AI systems, the latter topic being also the subject of the panel discussion organised by the cluster for the Hub's Annual Event.

5.4 Cluster 4: Encryption

Cluster leader: DG JRC

Cluster members: DG HOME, eu-LISA, Eurojust, Europol, FRA, GSC

The first priority of the Encryption Cluster was to contribute to the first report on encryption published by the EU Innovation Hub for Internal Security in June 2024. The cluster then had to await the recommendations of the High-Level Group on Access to Data for Effective Law Enforcement before holding its official kick-off meeting in September 2024. The cluster discussed which activities to prioritise, following up on the recommendations of the High-Level Group on Access to Data for Effective Law Enforcement. Also discussed were different initiatives in the area of quantum computing, in particular quantum-safe encryption, as well as testing and evaluation infrastructure for the European Quantum Communication Infrastructure (EuroQCI), which was identified as one of the possible areas of collaboration for cluster members in 2025.

6. Conclusion

This report presents the progress achieved by the EU Innovation Hub for Internal Security in 2024. The preliminary conclusion based on the results from 2024, and endorsed by the Steering Group at its November 2024 meeting, is that the Hub's new approach of working in thematic clusters has been successful. Each of the clusters has proven to be sustainable and will continue to operate in 2025. Moreover, a new cluster focusing on unmanned aerial systems and technologies countering their use has been proposed by Frontex and will kick off its operations in 2025.

In the absence of a dedicated budget for the Innovation Hub, having limited resources at its disposal and bearing in mind that agencies are not eligible to benefit from Horizon Europe funding, once again the work accomplished in 2024 by the Hub members should be considered as a significant achievement. The lessons learned through the delivery of the activities described above have been taken into consideration in the drafting of the Hub's work plan for 2025, which will be annexed to the multi-annual plan for 2023-2026.



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